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SUPERSEDING
FRANKFORD ARSENAL
PURCHASE DESCRIPTION
FA-PD-1828
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MILITARY SPECIFICATION

AMINOGUADINE BICARBONATE (AMINATE)

1. SCOPE

1.1 This specification covers aminoguanidine bicarbonate for use in the manufacture of tetracene.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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3. REQUIREMENTS

3.1 Material. The aminoguanidine bicarbonate shall be a white or cream powder.

3.2 Chemical properties. The chemical properties of aminoguanidine bicarbonate shall conform to the requirements shown in table I.

Table I. Chemical requirements

Requirements	Percent
Assay, min.	98.0
Acid-insoluble residue, max.	0.02
Moisture, max.	0.30
Ash, max.	0.10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Initial production sample. Prior to the initiation of a regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of three (3) pounds of aminoguanidine bicarbonate from the contractor's initial production, manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements herein specified. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the specification. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

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4.3 Size of lot. The quantity of aminoguanidine bicarbonate in a lot shall be as specified by the procuring agency.

4.3.1 Lot identification. Each lot of aminoguanidine bicarbonate shall be identified as to material, lot number, manufacturer, contract or order number, and number of pounds in the lot. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.4 Sampling. Three containers in each lot shall be selected in such a manner as to be representative of the lot. When lots comprise less than three containers, all the containers in the lot shall be selected. A representative sample of approximately 1/2 pound shall be removed from each container selected, by means of an instrument capable of collecting a dry sample.

4.5 Tests.

4.5.1 Preparation of reagents.

4.5.1.1 Ferrocyphen indicator solution. Dissolve 0.1 gram of "ferrocyphen" in 10 ml. of sulfuric acid. ("Ferrocyphen" is dicyano-bis-(1,10 phenanthroline)-iron II dihydrate and can be prepared as follows: Add 3 grams of 1,10 phenanthroline monohydrate and 1.95 grams of $FeSO_4 \cdot 7H_2O$ to 200 ml. of water and heat just below the boiling point. Add at one stroke 5 grams of potassium cyanide in 5 ml. of water. Stir occasionally for a few minutes on the hot plate and then allow to stand overnight at room temperature. Filter through a Buchner funnel, wash well with water, spread out on a watch glass, and allow to dry overnight at room temperature.)

4.5.1.2 Sulfanilic acid. Pulverize with a mortar and pestle, mix well, and place in a glass-stoppered bottle. Determine the purity by dissolving 0.5 gram (weighed to 0.1 mg) in 100 ml. of water and titrating with standard 0.1N sodium hydroxide solution to a phenolphthalein or bromthymol blue end point. Calculate the purity as follows:

$$\text{Percent sulfanilic acid (anhydrous)} = \frac{17.32 \times \text{ml of NaOH} \times \text{normality of NaOH}}{\text{grams of sample}}$$

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4.5.1.3 Standard sodium nitrite solution (0.5N). Dissolve 34.5 grams of reagent grade $NaNO_2$ in water and dilute to 1 liter. To standardize, transfer 2 grams of the sulfanilic acid (weighed to 0.1 mg) to a 400 ml. beaker. Dissolve in a mixture of 150 ml. of water and 100 ml. of hydrochloric acid by stirring with a stirring rod. Add 1 ml. of "ferrocyphen" indicator solution and proceed with the titration with sodium nitrite solution as described under 4.5.2. Calculate the normality of the sodium nitrite solution as follows:

$$\text{Normality of } NaNO_2 = \frac{0.05774 \times \text{grams of S.A.} \times \text{purity of S.A. in \%}}{\text{ml. of } NaNO_2}$$

4.5.2 Assay. Transfer 1.7 grams of the sample (weighed to 0.1 mg) to a 400 ml. beaker, add 150 ml. of water, and start magnetic stirring. Add 100 ml. of hydrochloric acid (the first 5-10 ml. slowly). Add 1 ml. of "ferrocyphen" indicator solution. Insert the tip of the burret below the surface of the solution and titrate with the standard sodium nitrite solution. Titrate fairly rapidly at first and dropwise near the end point. Allow 10-15 seconds between increments near the end point. The end point is a change from faint orange to a pure pale violet when viewed laterally through the vortex of the rapidly stirred solution against a white background. Calculate the percentage of aminoguanidine bicarbonate as follows:

$$\text{Percentage aminoguanidine bicarbonate} = \frac{13.61 \times \text{ml of } NaNO_2 \times \text{normality of } NaNO_2}{\text{grams of sample}}$$

4.5.3 Insolubility. Dissolve a 10 gram sample in a 250 ml. beaker with 50 ml. of water and 15 ml. of hydrochloric acid. Filter through a tared sintered glass crucible, transfer, and wash with water. Dry at 110°C for 1 hour, cool in a desiccator, and weigh. Calculate as follows:

$$\text{Percent insoluble matter} = \frac{\text{grams of residue}}{\text{grams of sample}} \times 100$$

Save the filtrate for the determination of sulfate.

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4.5.4 Moisture. Transfer a 10 gram sample to an open weighing dish (6 cm. in diameter), place in a desiccator containing fresh sulfuric acid, and allow to stand overnight. Determine the loss in weight. Calculate as follows:

$$\text{Percent moisture} = \frac{\text{grams loss in weight}}{\text{grams of sample}} \times 100$$

4.5.5 Ash. Transfer a 5 gram sample to a tared platinum crucible. Place the crucible on the hot plate at highest heat until the strong effervescence has ceased. Heat over a Meker burner until only small residue remains. Add 5 ml. of nitric acid, evaporate to dryness on the hot plate, and ignite for a few minutes over a Meker burner. Cool in a desiccator and weigh. Calculate as follows:

$$\text{Percent residue on ignition} = \frac{\text{grams of residue}}{\text{grams of sample}} \times 100$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Levels A and C. Aminoguanidine bicarbonate shall be packaged in accordance with the applicable requirements of PPP-C-301.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Levels A, B and C. Aminoguanidine bicarbonate shall be packed in accordance with the applicable requirements of PPP-C-301.

5.2 Marking. In addition to any special marking required by the contract or order, shipments, shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(1) Title, number and date of this specification.
(2) Level of packaging and packing required (see section 5).
(3) Unit quantity required.

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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